



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Software Construction And Development lab Report - 10

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PROGRAM: BSSE (5th Semester)

COURSE: Software Construction and Development

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DATE: 9/21/2025

DAY: Sunday

LAB 10 – Refactoring of Code

Task: Identify the smell in the code and refactor it.

Solution:

Identified Code Smells

- The main() method is too long.
- Switch cases contain repeated input code.
- Mixed responsibilities (input + logic together).
- No reusability.
- Hard to extend for new shapes.

```
import java.util.Scanner;

interface Shape {

    double calculateArea();

}

class Square implements Shape {

    private double side;

    public Square(double side) {

        this.side = side;

    }

    public double calculateArea() {

        return side * side;

    }

}

class Rectangle implements Shape {
```

```
private double length, width;
```

```
public Rectangle(double length, double width) {  
    this.length = length;  
    this.width = width;  
}
```

```
public double calculateArea() {  
    return length * width;  
}  
}
```

```
class Triangle implements Shape {  
    private double base, height;
```

```
public Triangle(double base, double height) {  
    this.base = base;  
    this.height = height;  
}
```

```
public double calculateArea() {  
    return 0.5 * base * height;  
}  
}
```

```
public class ShapeAreaCalculator {
```

```
public static void main(String[] args) {  
    Scanner scanner = new Scanner(System.in);
```

```
    int choice = getMenuChoice(scanner);
```

```

Shape shape = getShape(choice, scanner);

if (shape != null) {
    System.out.println("Area: " + shape.calculateArea());
} else {
    System.out.println("Invalid choice.");
}

scanner.close();
}

private static int getMenuChoice(Scanner scanner) {
    System.out.println("Select a shape:");
    System.out.println("1. Square");
    System.out.println("2. Rectangle");
    System.out.println("3. Triangle");
    return scanner.nextInt();
}

private static Shape getShape(int choice, Scanner scanner) {
    switch (choice) {
        case 1:
            System.out.print("Enter side: ");
            return new Square(scanner.nextDouble());

        case 2:
            System.out.print("Enter length: ");
            double l = scanner.nextDouble();
            System.out.print("Enter width: ");
            double w = scanner.nextDouble();
            return new Rectangle(l, w);
    }
}

```

case 3:

```
System.out.print("Enter base: ");
```

```
double b = scanner.nextDouble();
```

```
System.out.print("Enter height: ");
```

```
double h = scanner.nextDouble();
```

```
return new Triangle(b, h);
```

```
}
```

```
return null;
```

```
}
```

```
}
```